SQUEEZE THEOREM, MCT, INDUCTION

Theorem — **Squeeze Theorem**

If $a_n \leq b_n \leq c_n$ and $\lim_{n \to \infty} a_n = L = \lim_{n \to \infty} c_n$,

$\lim_{n \to \infty} b_n = L$.

<u>Proof</u>: Let $\varepsilon > 0$ be given. Since $a_n \to L$ and $c_n \to L$ there exists $N \in \mathbb{N}$ such that if $n \geq N$ then $a_n \in (L - \varepsilon, L + \varepsilon)$ and $c_n \in (L - \varepsilon, L + \varepsilon)$.

Then, for $N \in \mathbb{N}$, $L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon$, which implies $b_n \in (L - \varepsilon, L + \varepsilon)$.

Def'n — **Monoticity**

Given $\{a_n\}$ we define the following terms:

Requirement (for all n)	Name
$a_{n+1} > a_n$	increasing
$a_{n+1} \geq a_n$	non-decreasing
$a_{n+1} < a_n$	decreasing
$a_{n+1} \leq a_n$	non-increasing

Def'n

A sequence $\{a_n\}$ is <u>bounded above</u> if there exists a $B \in \mathbb{R}$ such that $a_n \leq B$ for all n.

A sequence $\{a_n\}$ is <u>bounded below</u> if there exists a $B \in \mathbb{R}$ such that $a_n \geq B$ for all n.

A sequence is <u>bounded</u> if it is bounded above and below.

Def'n

Given $\{a_n\}$, the <u>least upper bound (lub)</u> is a number M such that, $a_n \leq M$ and $M \leq B$ for any upper bound B.

Similarly, the <u>greatest lower bound (glb)</u> is a number m such that $a_n \geq m$ and $m \geq b$ for any lower bound b

Theorem — **Monotone Convergence Theorem**

If a sequence $\{a_n\}$ is monotonic, then

- it converges, if it is bounded
- it diverges, if it is unboun

Def'n

Mathematical Induction

Step 1 - Base Case
Step 2 - Inductive Hypothesis
Step 3 - Inductive Step

EX

Prove that $a_{n+1} = 5 - \frac{1}{a_n}$, $a_1 = 2$ satisfies $a_n \geq 1$

Let $P(n)$ denote the open sentence such that $a_n \geq 1$

<u>Base Case</u>: If $n = 1$,

$a_1 = 2$

Since $a_1 \geq 1$, $P(1)$ is true.

<u>Inductive Step</u>: Let $k \in \mathbb{N}$ be arbitrary. Suppose $P(k)$ is true, i.e. $a_k \geq 1$.

Consider $P(k+1)$.

$$\begin{aligned} a_{k+1} &= 5 - \frac{1}{a_k} \\ &\geq 5 - \frac{1}{1} \quad \text{by inductive hypothesis} \\ &= 4 \end{aligned}$$

Since $a_{k+1} \geq 1$, $P(k+1)$ is true. By POMI, $P(n)$ is true for all $n \in \mathbb{N}$.

QUIZ #3, PRACTICE PROBLEMS #3

PRACTICE PROBLEMS

1. In general, we have $a_n \sin(n) + b_n \cos(n) \leq a_n + b_n$ and $a_n \sin(n) + b_n \sin(n) \geq -a_n - b_n$

So, $-a_n - b_n \leq a_n \sin(n) + b_n \cos(n) \leq a_n + b_n$

Since $\lim_{n \to \infty} a_n + b_n = 0 = \lim_{n \to \infty} -a_n - b_n$,

$$\lim_{n \to \infty} a_n \sin(n) + b_n \cos(n) = 0$$

by the Squeeze Theorem.

2 (a) Consider $\{a_n\} = (-1)^n$ Then, it satisfies $\lim_{n \to \infty} |a_n| = L$ since $\lim_{n \to \infty} |(-1)^n| = 1$ but $\lim_{n \to \infty} (-1)^n$ DNE.

(b) By the def'n of a limit, for all $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that if $n \geq N$, $|a_n - L| < \varepsilon$.

Hence, $||a_n| - |L|| < \varepsilon$ if such n exists. Then,

$||a_n| - |L|| \leq |a_n - L|$ by given hint.

Since $|a_n - L| < \varepsilon \Rightarrow ||a_n| - |L| < \varepsilon$,

$$\lim_{n \to \infty} a_n = L \Rightarrow \lim_{n \to \infty} |a_n| = |L|$$

(c) The statement is true. Since $-|a_n| \leq a_n \leq |a_n|$ and $\lim_{n \to \infty} |a_n| = 0$, then $\lim_{n \to \infty} a_n = 0$ by the Squeeze Theorem.

3 (a) Note that $-1 \leq \sin(n^2) \leq 1 \Rightarrow \frac{-1}{n^2} \leq \frac{\sin(n^2)}{n^2} \leq \frac{1}{n^2}$

Since $\lim_{n \to \infty} \frac{-1}{n^2} = 0 = \lim_{n \to \infty} \frac{1}{n^2}$,

$\lim_{n \to \infty} \frac{\sin(n^2)}{n^2} = 0$ by the Squeeze Theorem.

(b) Note that $-1 \le (-1)^n \le 1$ Then,

$$\frac{3n+1}{n} \le \frac{3n-(-1)^n}{n} \le \frac{3n-1}{n}$$

Since $\lim\limits_{n\to\infty} \dfrac{3n/n + 1/n}{n/n} = \lim\limits_{n\to\infty} 3 + 0 = 3$ and

$\lim\limits_{n\to\infty} \dfrac{3n/n - 1/n}{n/n} = \lim\limits_{n\to\infty} 3 + 0 = 3$,

$\lim\limits_{n\to\infty} \dfrac{3n-(-1)^n}{n} = 3$ by the Squeeze Theorem.

(c) Consider $\dfrac{n!}{n^n}$ as $\dfrac{1\cdot 2\cdot 3\cdot \ldots \cdot n}{n\cdot n\cdot n\cdot \ldots\ n}$

Note that $0 < \dfrac{n!}{n^n} \le 1 \Rightarrow \dfrac{0}{n^n} < \dfrac{n!}{n^n} \le \dfrac{1}{n^n}$.

Since $\lim\limits_{n\to\infty} 0 = 0 = \lim\limits_{n\to\infty} \dfrac{1}{n^n}$,

$\lim\limits_{n\to\infty} \dfrac{n!}{n^n} = 0$ by the Squeeze Theorem.

(d)

$$= \lim_{n\to\infty} \frac{3n^3/n^3 + 2n^2/n^3 - n/n^3 - 1/n^3}{n^3/n^3 + n/n^3 + 3/n^3}$$

$$= \lim_{n\to\infty} \frac{3 + 1/2n - 1/n^2 - 1/n^3}{1 + 1/n^2 + 3/n^3}$$

$$= \frac{3+0+0+0}{1+0+0}$$

$$= 3$$

(e)

$$= \lim_{n\to\infty} \frac{n^2/n + 2n/n - 6/n}{n/n + 1/n}$$

$$= \lim_{n\to\infty} \frac{n + 2 - 6/n}{1 + 1/n}$$

$$= \frac{\infty + 2 + 0}{1 + 0}$$

$$= \infty$$

4 (a)

<u>Base Case</u>: $a_1 = 1 < a_2 = \frac{4}{3}$ as required.

<u>Inductive Hypothesis</u>: Suppose $a_k < a_{k+1}$ for some $k \in \mathbb{N}$. Then,

$$a_k < a_{k+1} \Rightarrow 7 + a_k < 7 + a_{k+1} \Rightarrow$$

$$\frac{7 + a_k}{6} < \frac{7 + a_{k+1}}{6} \Rightarrow a_{k+1} < a_{k+2} \text{ as required}$$

Thus, by POMI, $\{a_n\}$ is an increasing sequence for $n \geq 1$

<u>Base Case</u>: $a_1 = 1 \leq 2$ as required.

<u>Inductive Hypothesis</u>: Suppose $a_k \leq 2$ for some $k \in \mathbb{N}$. Then,

$$a_{k+1} \leq 2 \Rightarrow 7 + a_k \leq 9 \Rightarrow \frac{7 + a_k}{6} \leq \frac{3}{2} \leq 2 \Rightarrow$$
$$a_{k+1} \leq 2 \text{ as required}$$

Thus, by POMI, $\{a_n\}$ is bounded above by 2 for all $n \geq 1$

(b)

Since $\{a_n\}$ is an increasing sequence and bounded above by 2, it is convergent by MCT. Hence $\{a_n\}$ has a limit, call it L. Then,

$$L = \lim_{n \to \infty} a_n = \lim_{n \to \infty} a_{n+1} = \lim_{n \to \infty} \frac{7 + a_n}{6} = \frac{7 + L}{6}$$

$$L = \frac{7 + L}{6} \Rightarrow 6L = 7 + L \Rightarrow 5L = 7 \Rightarrow L = \frac{7}{5}$$

Thus, $\lim_{n \to \infty} a_n = \frac{7}{5}$

5 (a) <u>Base Case</u>: $a_1 = \sqrt{2} < a_2 = \sqrt{2+\sqrt{2}}$ as required.

<u>Inductive Hypothesis</u>: Suppose $a_k < a_{k+1}$ for some $k \in \mathbb{N}$. Then,

$$a_k < a_{k+1} \Rightarrow 2 + a_k < 2 + a_{k+1} \Rightarrow$$

$$\sqrt{2+a_k} < \sqrt{2+a_{k+1}} \Rightarrow a_{k+1} < a_{k+2}$$

Thus, by POMI, $\{a_n\}$ is an increasing sequence.

<u>Base Case</u>: $0 \leq a_1 = \sqrt{2} \leq 3$ as required.

<u>Inductive Hypothesis</u>: Suppose $0 < a_k \leq 3$ for some $k \in \mathbb{N}$ Then,

$$0 \leq a_k \leq 3 \Rightarrow 2 \leq a_k \leq 5 \Rightarrow \sqrt{2} \leq \sqrt{2+a_k} \leq \sqrt{5} \leq 3$$
$$\Rightarrow 0 \leq a_{k+1} \leq 3 \text{ as required.}$$

Thus, by POMI, a_k is bounded below by 0 and above by 3.

(b) Since $\{a_n\}$ is an increasing sequence and bounded, it is convergent by MCT. Hence, $\{a_n\}$ has a limit, call it L.

$$L = \lim_{n\to\infty} a_n = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \sqrt{2+a_n} = \sqrt{2+L}$$

$$L = \sqrt{2+L} \Rightarrow L^2 = 2 + L \Rightarrow L^2 - L - 2 = 0.$$

Thus, $L = -1$ or $L = 2$ But since $\{a_n\}$ is bounded below by 0, $L \neq -1$. Hence, $\lim_{n\to\infty} a_n = 2$.

6 (a) T
(b) F
(c) F
(d) T
(e) T
(f) T

7. (a) T
(b) F
(c) T
(d) T